CCC-C-428c
October 22, 1962

SUPERSEDING
Int. Fed. Spec. CCC-C-00428b (ARMY-QMC)
June 5, 1962 and
Fed. Spec. CCC-C-428a
October 27, 1959

FEDERAL SPECIFICATION

CLOTH, COTTON, DUCK; FIRE, WATER, WEATHER, AND MILDEW RESISTANT

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for cloth, cotton, duck; fire, water, weather and mildew resistant treated.

1.2 Classification.

1.2.1 *Types and classes.* The finished cloth shall be of the following types and classes as specified (see 6.2):

Type I. — Cloth finished in olive drab No. 7 color.

Type II. — Cloth finished in deck grey color.

Class 1. — Regular finish.

Class 2. — Dry finish.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

CCC-C-419 — Cloth, Cotton, Duck, Unbleached, Plied-Yarns (Army and Numbered).

CCC-C-443 — Cloth, Cotton, Duck; (Single and Plied Filling Yarns Flat).

CCC-T-191 — Textile Test Methods.

PPP-P-51 — Packaging, Packing, and Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks and Synthetics).

Federal Standards:

Fed. Std. No. 4 — Glossary of Fabric Imperfections.

Fed. Std. No. 102 — Preservation, Packaging and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-C-2384 — Cloth, Cotton, Duck, Plied and Single Yarns (High Sley).

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Military Standard:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Rules and Regulations Under the Textile Fiber Products Identification Act, Effective March 3, 1960.

(Copies may be obtained without charge from the Federal Trade Commission, Washington 25, D.C.)

Munsell Book of Color

(Copies may be obtained from the Munsell Color Company, 10 East Franklin Street, Baltimore, Maryland).

3. REQUIREMENTS

3.1 Standard sample. When a standard sample is available, the cloth shall match the standard sample for shade and shall be equal to, or better than, the standard sample with respect to all characteristics for which the standard sample is referenced. (see 6.4).

3.2 Laboratory report approval and bid sample approval. Unless otherwise specified at the time of submission of a bid, the prospective contractor shall submit to the contracting officer for approval a sample 5 by 9 inches (minimum) and a certified copy of a recent laboratory report covering the type (i.e. flat duck, Army duck, numbered duck, etc.) which he proposes to deliver (see 6.2). Unless otherwise specified by the contracting officer, the bidder shall certify that the sample and laboratory report submitted cover cloth which was finished in a mill where the finishing will be performed if the contract is awarded. The laboratory report shall contain test data which demonstrate that the finished cloth, which the contractor proposes to deliver, has been tested and has been found to comply with all requirements of this specification. Any of the following types of reports will be satisfactory from the standpoint of this requirement:

(a) An independent or commercial laboratory report.

(b) The prospective bidder's own laboratory report.

(c) A Government laboratory report from a contract within not more than 6 months of date of submission of bid.

The purpose of the above requirement is to assist the Government to determine the capability of bidders to manufacture a cloth meeting all the requirements of the specification. The submission of an acceptable sample under this requirement shall not be construed as relieving a contractor from subsequently meeting all requirements of the specification on all deliveries.

3.3 Formula approval. The prospective contractor or his finishing plant shall submit to the contracting officer for approval a list of materials which the prospective contractor or his finishing plant intends to use, stipulating grades and the standards of all materials including solvents. This list is for the information of the Government only and is not to be made available to the public (see 6.7). The use of compounds containing mercury in any form is prohibited.

3.3.1 *Disapproval of formula.* In the event that the list of proposed materials is found to contain products which past experience has indicated may detract from the quality of the performance or which may present a toxic hazard to canvas-goods workers, the proposed formulation shall be rejected (see 6.7).

3.3.2 *Approval of formula.* Approval of the qualitative formula submitted under 3.3

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shall in no way absolve the contractor from complying with the performance requirements of this specification.

3.4 Materials.

3.4.1 *Basic cloth.* The cloth shall be as specified by the contracting agency and shall conform to CCC-C-419, CCC-C-443 or MIL-C-2384 (see 6.2). Sized cloth prior to finishing shall contain not more than 2.5 percent starch and protein materials and not more than 5.0 percent total size, finishing, and other non-fibrous materials including starch and protein when tested as specified in 4.4.

3.4.2 *Grey-cloth weight.* Unless otherwise specified in subsequent tables and paragraphs where reference is made to grey-cloth weight, the minimum acceptable grey-cloth weight as given in the applicable grey-cloth specification shall be used to determine the requirements under these paragraphs.

3.4.3 *Compound.* The compound to provide fire, water, weather, and mildew resistance shall be well ground and evenly blended. Compounds containing sulfur may be permitted when specifically indicated by name in formulae approved by the contracting officer. The compound shall be of the same ingredients as contained in the list of materials submitted and approved under 3.3. Compounds containing mercury in any form are prohibited.

3.5 Finished cloth.

3.5.1 *Construction.* The number of yarns per inch of the finished cloth shall conform to the requirements of the specification covering the grey cloth specified except that a minus tolerance in the filling yarn will be allowed as follows:

Designation	Minus tolerance
Flat ducks (requiring desizing)	Up to 10 percent
Army duck	Up to 5 percent
No. 8 and lighter duck	Up to 5 percent
Heavier than No. 8 duck	Up to 3 percent

3.5.2 *Finished width.* The width of finished cloth shall be as specified in the contract or order (see 6.2 and 6.8) with the following tolerances permitted:

Numbered duck

Width of duck, Inches	*Tolerance, Inch*
40 or less	— ½ + ¼
Over 40 to 51 incl.	— ½ + ⅜
Over 51 to 81 incl.	— ¾ + ½
Over 81	— ⅞ + ⅝

Army duck

Width of duck, Inches	*Tolerance, Inch*
40 or less	— ¼ + ½
Over 40 to 51 incl.	— ¼ + ⅝
Over 51 to 81 incl.	— ⅜ + ¾
Over 81	— ½ + ⅞

Flat and high sley ducks—up to 15 ounces
(Requiring desizing)

Width of duck, Inches	*Tolerance, Inch*
40 or less	— 1¾ + ⅜
Over 40 to 51 incl.	— 2¼ + ½
Over 51 to 81 incl.	— 2⅝ + ⅝
Over 81	— 2⅞ + ¾

Flat and high sley ducks—over 15 ounces
(Requiring desizing)

Width of duck, Inches	*Tolerance, Inch*
40 or less	— 1½ + ⅜
Over 40 to 51 incl.	— 1¾ + ½
Over 51 to 81 incl.	— 2¾ + ⅝
Over 81	— 2⅝ + ¾

3.5.3 *Breaking strength.* The breaking strength of the finished cloth in both warp and filling directions shall be not less than that of the corresponding specification requirements for the grey cloth.

3.5.4 *Weight.* The weight of the finished cloth shall be the actual weight of the grey cloth plus an allowance for the treatment in accordance with table I.

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TABLE I

Grey weight (ounces per sq. yd.)	Allowable maximum increase in weight due to treatment based on actual grey cloth weight (Percent)
12 and over	50
Under 12	65

3.5.5 *Color.* The color shall be as specified for the applicable type and shall match the standard sample (see 6.2) when furnished by the contracting activity (see 6.4). The color shall be obtained with carbon black or lamp black, inorganic or phthalocyanine pigments.

3.5.5.1 *Colorfastness.* When a standard sample is available, the colored and finished cloth shall show fastness to accelerated weathering equal to, or better than, the standard sample after 100 hours when tested as specified in 4.4. When a standard sample is not available or is not referenced as a standard for colorfastness, the colored and finished cloth shall show "good" colorfastness to accelerated weathering after 100 hours when tested as specified in 4.4.

3.5.5.2 *Matching.* The color of the finished cloth shall match the standard shade under a natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.5.6 *Crocking fastness.*

3.5.6.1 *Class* 1. Cloths with the regular finish shall show no more crocking on dry white cotton cloth than the Munsell neutral value of 5.0 when tested as specified in 4.4.

3.5.6.2 *Class* 2. Cloths finished with a dry finish for upholstery purposes shall show no more crocking on dry white cotton cloth than Munsell neutral value of 7.0 when tested as specified in 4.4.

3.5.7 *Flexibility.* When tested in accordance with procedures specified in 4.4, the finished cloth shall show a bending moment of not more than that specified in table II for the given untreated numbered duck or Army duck classified in CCC-C-419, where applicable.

TABLE II. *Flexibility*

CCC-C-419 Cloth designations	Initial bending moment inch-lbs. maximum	After heating at 200°-205° F. (bending moment—inch-lbs.) maximum	At 0° ± 5° F. (bending moment—inch-lbs.) maximum
Numbered			
2	0.16	0.400	0.400
4	.12	.300	.300
6	.06	.180	.180
8	.05	.125	.125
10	.03	.075	.075
12	.02	.050	.050
Army ducks			
8.25	0.013	0.032	0.032
9.85	.013	.032	.032
12.29	.013	.032	.032

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3.5.7.1 For all other cloths, the flexibility requirements shall be as specified (see 6.2).

3.5.8 *Water resistance.* The finished cloth shall have a maximum leakage of 50 ml. at the height indicated in table III when tested by the water-permeability test specified in 4.4. Where finishing according to this specification is to be contracted for by using Government furnished grey cloth, adjustment shall be made on the water permeability, such that the finished cloth may exceed the 50 ml. requirement but not to exceed 25 percent of the grey cloth for water permeability when tested as specified in 4.4. The cloth shall conform to the same maximum leakage requirements after the accelerated weathering test specified in 4.4.

3.5.8.1 The requirements for resistance to water permeability shall be applicable only to the flat ducks of CCC-C-443 having weights of 14.90 and 17.55 ounces per square yard and to all ducks of CCC-C-419 and MIL-C-2384.

TABLE III. *Height for test of water resistance*

Untreated weight (oz/sq/yd)	Hydrostatic height	
	Inches	*Centimeters*
Over 20	20	50.80
14 to 20	14	35.56
Under 14	8	20.32

3.5.9 *Fire resistance.* When tested as specified in 4.4, the average time of flaming of the specimens tested for each sample unit shall not exceed 2 seconds after the burner flame is withdrawn, and no specimen shall in any case exceed 4 seconds of flaming after the burner flame is withdrawn. The average length of char shall not exceed the requirements of table IV. The cloth shall conform to the same fire-resistance requirements after the 100-hour accelerated weathering period as specified in 4.4 and also after being exposed to water leaching when tested as specified in 4.4.

TABLE IV. *Average length of char*

Untreated weight (oz/sq/yd)	Maximum average length of char (inches)
Under 10	4.5
10 to 20	3.5
Over 20	2.0

3.5.10 *Mildew resistance.*

3.5.10.1 *For type I cloths.* The cloth shall be made mildew resistant by proper deposition within the cloth of a fungicide which is currently approved for this use by the procuring activity. Unless otherwise specified (see 6.2), the fungicide shall be one of the following in the specified concentration:

(*a*) Copper-8-quinolinolate to deposit a minimum of 0.18 percent copper as metal from copper-8-quinolinolate to a maximum of 0.27 percent copper as metal from copper-8-quinolinolate.

(*b*) Copper naphthenate combined with copper-8-quinolinolate to deposit a minimum of 0.35 percent total copper with a minimum of 0.05 percent copper as metal from copper-8-quinolinolate to a maximum of 1.48 percent total copper with a maximum of 0.18 percent copper as metal from copper-8-quinolinolate.

(*c*) Copper-hydroxy naphthenate combined with copper-8-quinolinolate to deposit a minimum of 0.40 percent total copper with a minimum of 0.05 percent copper as metal from copper-8-quinolinolate to a maximum of 1.48 percent total copper with a maximum of 0.18 percent copper as metal from copper-8-quinolinolate.

The fungicide shall be uniformly applied and the cloth shall be well penetrated by the

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fungicide so that specimens leached as specified in 4.4 shall show no more than traces of any organism when observed with a hand lens of approximately 3X magnification after being tested as specified in 4.4. Total copper content shall be determined as specified in 4.4.

3.5.10.2 *For type II cloths.* The cloth shall be made mildew resistant by proper deposition within the cloth of a fungicide which is currently approved for this use by the procuring activity. Two specimens shall be evaluated. The first specimen shall be the treated cloth as received and the second specimen shall be the treated cloth after the accelerated weathering test followed by the leaching resistance test as specified in 4.4. Neither specimen shall show more than traces of mildew when tested by the method specified in 4.4. The treated cloth shall not lose more than 15 percent of its strength due to mildew attack after the specimen has been tested as received, and after accelerated weathering followed by the leaching resistance test, when tested as specified in 4.4.

3.5.11 *Resistance to accelerated weathering.*

3.5.11.1 *Army and numbered duck.* When a standard sample for weathering is available, the finished cloth shall be equal to or better than the standard sample when exposed to 100 hours accelerated weathering and tested immediately as specified in 4.4. When a standard sample is not available, the finished cloth, after exposure to 100 hours accelerated weathering as specified in 4.4 shall conform to the following requirements:

Untreated (grey) weight (ounces per square yard)	Minimum percent retention of finished fabric breaking strength
15 oz. and over	60 percent
Under 15 oz.	75 percent

3.5.11.2 *Alternate Army ducks.* The breaking strength requirements for the material listed herein shall be the same as specified in 3.5.11.1 for Army and numbered duck:

MIL-C-2384 — Type I, Class 1, 10.75 ounce (High sley)
— Type I, Class 2, 14.75 ounce (High sley)

CCC-C-443 — Type I, 14.90 ounce (Flat duck)
— Type I, 17.55 ounce (Flat duck)

3.5.11.3 *Flat ducks.* The breaking strength for flat ducks except those listed in 3.5.11.2 shall be not less than 80 percent of the breaking strength of the finished cloth, or not less than that specified in the base-cloth specification after exposure to 200 hours accelerated weathering when tested as specified in 4.4.

3.6 Fiber identification. Each roll shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations Under The Textile Fiber Products Identification Act, effective March 3, 1960 (see 2.2).

3.7 Put-up. The cloth shall be put up on rolls in accordance with PPP-P-51.

3.8 Workmanship. The cloth shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the acceptable quality levels.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Contractor's inspection responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the

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inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Pre-acceptance inspection.

4.2.1 *Laboratory report approval.* The data contained in the report submitted in accordance with 3.2 shall be obtained in accordance with test methods specified in 4.4.

4.3 Inspection for acceptance. Except when otherwise specified by Civil Government agencies, inspection shall be in accordance with the provisions of MIL-STD-105, except where otherwise indicated.

4.3.1 *Examination of the end item.* The cloth shall be examined to determine conformance with the requirements of this specification. Defects found during this examination shall be classified in accordance with 4.3.1.1, 4.3.1.2 and 4.3.1.3.

4.3.1.1 *Yard-by-yard examination.* The defects listed in tables V(A) and V(B) as applicable shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The sample unit for this examination shall be 1 linear yard. The lot size shall be expressed in units of 1 yard each. The sample size shall be in accordance with inspection level II of MIL-STD-105. The acceptable quality levels for this examination shall be as follows:

(a) Finished cloth when the grey cloth is furnished by the Government (table V(B))

(1) Width of cloth 56 inches or less: 0.25 major defects and 1.5 total defects (major and minor combined) per 100 units (yards).

(2) Width of cloth more than 56 inches: 0.40 major defects and 2.5 total defects (major and minor combined) per 100 units (yards).

(b) Finished cloth when grey cloth is contractor furnished (table V(A))

(1) Width of cloth 56 inches or less: 0.40 major defects and 2.5 total defects (major and minor combined) per 100 units (yards).

(2) Width of cloth more than 56 inches: 0.65 major defects and 4.0 total defects (major and minor combined) per 100 units (yards).

4.3.1.2 *Overall examination.* Applicable when grey duck is supplied either by the Government or by the contractor. Each defect listed below shall be counted no more than once in each piece. The sample size and acceptance number shall be as shown in table V(C).

Defects

Baggy, ridgy or wavy cloth.

Color not uniform, off shade (not within established tolerances), shaded end to end, side to side, or side to center.

Poorly constructed — not firmly or tightly woven.

Width not within specified tolerance.

Overall application of compound not uniform.

Objectionable odor.

Overall uncleanliness.

Clearly noticeable crystallization of mildew inhibitor.

Tackiness, streaks or areas of color other than the designated shade. Any removal of surface color (by hand rubbing or pressure) resulting in a shade change other than that designated.

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TABLE V(A). *Applicable to finished cloth when grey cloths are supplied by the contractor*

Glossary Number Fed. Std. No. 4	Defect	Classification	
		Major	Minor
55, 56, 90, 91, 92	Hole, cut, tear or break (including selvage)—any	X	
78	Smash—any	X	
79, 89	Abrasion mark, bruise, tender or weak spot—any	X	
11, 33	Broken or missing yarns: Three or more contiguous regardless of length	X	
	Two contiguous, 2 inches or more missing	X	
	Two contiguous, less than 2 inches or single 2 inches or more missing		X
9, 12, 35	Fine or light filling bar, open place or crack—clearly visible[1]	X	
2	Coarse filling bar or heavy place clearly visible[1] and exceeding 2 linear inches		X
5, 26, 27, 77	Slough off, jerked in filling, slubs, gouts, knots—more than three times the thickness of normal yarns		X
13, 34	Floats or skips—multiple ½ inch or more in combined warp and filling directions, or harness skips	X	
	Multiple less than ½ inch in combined warp and filling directions, or single 2 inches or more in length		X
41	Reed mark or temple mark—resulting in a clear warpwise separation markedly wider than normal		X
40	Wrong draw—½ yard or more		X
49, 50	Slack or tight selvage resulting in pronounced slackness or waviness in cloth		X
51	Scalloped selvage—indentations, and exceeding ½ inch		X
53, 54	Stringy, beaded, corded, loopy or rough selvages ½ yard or more, resulting in raised places along the selvage edge 2 or more times the normal cloth thickness, ½ yard or more		X
74, 75, 76[2]	Spot, strain or streak 1 inch or more in combined directions—clearly visible[1]	X	
	More than ¼ inch but less than 1 inch in combined directions clearly visible[1]		X
48, 60	Selvage curled or folded, or crease-embedded, one half yard or more		X
57, 59, 61	Wrinkle, crease or finishing mark	X	

[1]At normal inspection distance (approximately 3 feet).

[2]Except counting of defects of spots, stains or streaks shall not be restricted to rust, oil or grease.

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TABLE V(B). *Applicable to finished cloth when grey cloth is either Government or contractor furnished*

Examine	Defect	Classification	
		Major	Minor
Hole, cut, tear or weak place (including selvage) caused in finishing operation	Clearly visible[1]	X	
Selvage curled or folded or crease-embedded	One half yard or more		X
Wrinkle, crease or finishing mark	1 inch or more in combined directions clearly visible[1]	X	
Spots, stains, streaks, shade bar (caused in finishing process)	More than ¼ inch but less than 1 inch in combined directions—clearly visible[1]		X

[1]At normal inspection distance (approximately 3 feet).

TABLE V(C). *Sample size*

Lot size in yards	Sample size in rolls	Maximum number defects acceptable in sample
Up to 1,300[1]	3	0
1,301 to 3,200	5	0
3,201 to 8,000	7	0
8,001 to 22,000	10	0
22,001 to 110,000	15	0
110,001 and over	25	1

[1]If a lot contains fewer than 3 rolls, each roll in the lot shall be examined.

4.3.1.3 *Length examination.* Applicable when grey duck is supplied either by the Government or by the contractor.

4.3.1.3.1 *Examination for length of individual rolls* Each roll in the sample shall be examined for gross length. Any gross length found to be less than the specified minimum length or more than the maximum gross length permitted, or any gross length found to be more than 2 yards less than the gross length marked on the ticket, shall be considered a defect with respect to length. The sample unit for this examination shall be one roll. The sample size and acceptance number shall be as shown in table V(C).

4.3.1.3.2 *Examination for total yardage in sample.* The lot shall be unacceptable if the total of the actual gross lengths of the rolls in the sample is less than the total of the gross lengths marked on the piece tickets. The rolls examined shall be those selected for examination of individual rolls.

4.3.2 *Examination of preparation for delivery requirements.* An examination shall be made to determine that packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping roll fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping rolls fully prepared for delivery. The lot size shall be the number of shipping rolls in the end item inspection lot. The inspection level shall be L-4 and the AQL shall be 4.0 defects per 100 units.

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Examine	Defects
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing. Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components, such as: incomplete closure of rolls, inadequate securing or sealing of wrapping materials, insecure sewing or loose wire ties.
Weight	Gross weight exceeds requirements.

4.4 **Testing of the end item.** The physical and chemical values specified in section 3 (except where otherwise indicated) apply to the average of the determinations made on a sample unit for test purposes as specified in CCC-T-191, wherever applicable and as listed in table VI shall be followed. The tests shall be subdivided into two categories as follows:

(a) Short term tests. Except when otherwise specified by Civil Government agencies, tests to be performed on a lot by lot basis. These tests are designated by a single asterisk in table VI. The acceptable quality level for each characteristic shall be 6.5 test failures per 100 units. The sample size shall be in accordance with inspection level L-1 of MIL-STD-105. The lot size shall be expressed in units of 1 linear yard each. The sample unit for testing purposes shall be 3-1/2 yards of finished cloth and 1-1/2 yards of grey cloth.

(b) Long term tests. Tests to be performed in accordance with table VI. These tests are designated by a double asterisk in table VI. Except when otherwise specified by Civil Government agencies, the sample size shall be in accordance with inspection level L-1 of MIL-STD-105. Long term tests shall be performed on the basis of the following frequency:

1. On the first lot.

2. Once in every five lots thereafter.

The lot size shall be expressed in units of 1 linear yard each. The sample unit for testing purposes shall be 2-1/2 yards of finished cloth and 1/2 yard of grey cloth.

Failure to meet a requirement for a long term test will be cause for rejection of the lot in question and will result in the characteristic for which the failure occurred being placed on a lot-by-lot basis for test purposes until three consecutive lots shall have passed the requirement.

When the data in the "number of determinations per individual sample units" and "results reported as" columns are not specified in the following table, they shall be as required by the referenced test method.

4.4.1 *Conditioning of samples.* All samples shall be conditioned for at least 16 hours in an atmosphere having a relative humidity of 65 ± 2 percent and a temperature of 70° ±2°F. before testing.

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TABLE VI. *Test method*

*Short term test (see 4.4.(a))
**Long term test (see 4.4.(b))

Characteristics	Req. ref.	Method No. and para. ref.	No. det. per individual sample unit	Results reported as:
Grey cloth:				
Weight*	3.4.2	5041	5	Average of 5 determinations to nearest 0.01 ounce
Non-fibrous materials content 5*	3.4.1	2611		
Materials:				
Compound*	3.4.3[8]	Std. commercial	1	Pass or fail
Finished cloth:			5	
Weight*	3.5.4	5041		Average of 5 determinations to nearest 0.01 ounce
Yarns per inch*	3.5.1	5050		
Breaking strength:				
Initial*	3.5.3	5100		
After accel. wea.**	3.5.11.1 3.5.11.2 3.5.11.3	5100,5804 and 4.4.3		
Flexibility:				
Initial*	3.5.7	5202, 4.4.2	8	Average of 4 determinations to nearest 0.001 inch lb. for both warp and filling
After heat aging**	3.5.7	5202, 4.4.2	8	
At low temperature**	3.5.7	5202, 4.4.2	8	
Color*	3.5.5	Visual	1	Satisfactory or unsatisfactory
Colorfastness to:				
Accel. weather**	3.5.5.1	5671, 4.4.3[2]		
Crocking*	3.5.6	5651		
Flame resistance:				
Initial:*				
Char length	3.5.9	5903	8	Average of 4 determinations to nearest 0.1 inch char for both warp and filling
Flaming time	3.5.9	5903	8	Each determination and average of 4 determinations to nearest 1/5 second for warp and filling
After accel. wea.:**				
Char length	3.5.9	5903, 4.4.3	8	Average of 4 determinations to nearest 0.1 inch char for both warp and filling.

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TABLE VI. *Test methods* (Cont'd)

Characteristics	Req. ref.	Method No. and para. ref.	No. Det. per individual sample unit	Results reported as:
Flaming time	3.5.9	5903, 4.4.3	8	Each determination and average of 4 determinations to nearest 1/5 second for each of the warp and filling.
After water leaching:[3]**				
Char length	3.5.9	5903, 5832	8	Average of 4 determinations to nearest 0.1 inch char for both warp and filling
Flaming time	3.5.9	5903, 5832	8	Each determination and average of 4 determinations to nearest 1/5 second, for each of the warp and filling.
Dye identification[8]*	3.5.5	Std. comml.	1	Pass or fail
Water resistance:[4]				
Grey cloth*	3.5.8	5516		
Treated cloth:				
Initial*	3.5.8	5516		
After accel. weather**	3.5.8	5516-5804 and 4.4.3		
Mildew resistance for:				
Type I:*				
Total copper analysis [6-7]	3.5.10.1	2050 or 2051		
Copper-8-quinolinolate[7]	3.5.10.1	2050 or 2051		
Type I**	3.5.10.1	5832, 5760	1	Growth or no growth
Type II:**				
1. As received:				
A. Growth	3.5.10.2	5760		
B. Loss in tensile strength	3.5.10.2	5760		
2. After accel. wea. and leaching:				
A. Growth	3.5.10.2	5760, 5804 and 5832		
B. Loss in tensile strength	3.5.10.2	5760, 5804 and 5832		

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[1]Method numbers refer to CCC-T-191, paragraph and table references apply to this specification.

[2]The test apparatus shall be as specified in 4.4.3.1 except that the filters shall not be removed.

[3]Leaching period shall be 24 hours.

[4]Method 5516 shall be used except that the specimen shall be immersed in water at a temperature 75° to 85° F. to a depth of 2 inches for 2 hours and shall be tested while still wet. The hydrostatic head for testing shall be as specified in table III of this specification and time of exposure to this hydrostatic head shall be 10 minutes.

[5]The amount of non-fibrous materials shall be determined with following formula:

$$\text{Total non-fibrous material, percent} = \frac{\text{O-S}}{\text{O}} \times 100$$

$$\text{Starch and protein material, percent} = \frac{\text{O-S-W-C}}{\text{O}} \times 100$$

[6]Contractors shall indicate on their test reports which fungicides were used.

[7]The contractor shall certify that required amount of mildew inhibitors in the combination specified have been used in the treatment of duck.

[8]This requirement may be accepted on the basis of contractor's certification of compliance with specification requirements in lieu of test results.

4.4.2 *Flexibility tests.* The test shall be conducted in accordance with Method 5202 with the following modifications:

Apparatus.—The bending moment required to deflect a 1-inch strip, ½ inch span, to a 20 degree deflection shall be measured. The rate of angular deflection shall be 60 degrees per minute.

Specimens.—Eight specimens of the conditioned sample (see 4.4.1) 1 inch by 2 inches shall be cut, four with the long dimension in the warp direction and four in the filling direction, and pressed between two 6- by 6-inch glass plates weighted with a 20-pound weight for four hours at standard conditions.

4.4.2.1 *After heat aging.* The specimen shall be tested for flexibility after being subjected to a temperature between 200° and 205°F. for 120 hours in a well-ventilated oven and then being allowed to cool at standard conditions for 16 hours.

4.4.2.2 *At low temperature.* Flexibility test shall be conducted at a temperature of 0° ± 5° F. upon specimens which have been subjected to that temperature for not less than one hour.

4.4.3 *Accelerated weathering test.*

4.4.3.1 *Apparatus.* Unless otherwise specified, the apparatus shall be as specified in method 5804, except that:

(a) Delete from title "(National Weathering Unit)".

(b) The filters shall be removed when testing the following materials:

CCC-C-419—All types Army and numbered duck.

MIL-C-2384—Type I, class 1, 10.75 ounce (high sley)
—Type I, class 2, 14.75 ounce (high sley)

MIL-C-443 —Type I, 14.90 ounce (flat duck)
—Type I, 17.55 ounce (flat duck)

(c) Water spray nozzles shall be mounted horizontally (the water spray assembly vertically) in the test chamber inside the specimen rack and so placed that the water shall strike the specimens evenly over their

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entire length in the form of a fine continuous spray in sufficient volume to cover specimens immediately on impact. The apparatus shall be so operated that the specimens are exposed to successive cycles of 102 minutes of light without spray, and 18 minutes of light with spray.

(d) The rack shall rotate about the arc at a uniform speed of one revolution per minute.

(e) The pressure of the water entering the spray shall be 12-18 p.s.i inclusive.

(f) The black panel thermometer shall be mounted on the test panel rack and readings taken at a point where maximum heat is developed and where water spray is not striking the panel. The black panel temperature at the exposure plane of the specimen rack shall be 155° plus or minus 10° F. when the filters are in place and 175° plus or minus 10° F. when the filters are removed.

(g) Note: Attention is called to manufacturers who own equipment which operates at one cycle in two hours. This equipment may continue to be used. However the one cycle per minute shall be the standard procedure in the event of dispute.

4.4.3.2 *Procedure.* Two specimens, 9-1/2 by 28 inches, shall be cut from the treated cloth under test. One specimen shall be cut with the long dimension in the direction of the warp; the other with the long dimension in the direction of the filling. In the event that the width of the treated cloth does not permit a full 28-inch specimen to be taken in the filling direction, two 9-1/2- by 14-inch specimens shall be cut. The specimens shall be exposed one above the other in the quadrant of the accelerated weathering apparatus. The specimens shall be changed from the top to bottom racks approximately every 25 hours during the designated exposure period (see 3.5.11). At the conclusion of the exposure period, breaking strength and fire resistance specimens shall be cut from the exposed material. The four fire resistance specimens in each direction of the treated cloth shall be exposed to accelerated weathering in a vertical position. In conducting fire resistance tests on these specimens, two specimens shall be subjected to the test flame at the end which was at the top of the exposed material, and two shall be subjected to the test flame at the end which was at the bottom of the specimen. The lower edges of the fire resistance test specimens shall be trimmed if necessary so that a freshly cut end is exposed to the test flame.

4.4.3.3 The specimens shall be exposed for the designated period, after which they shall be allowed to dry, and then be conditioned at standard conditions before testing.

5. PREPARATION FOR DELIVERY

(Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civilian procurements.)

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2 and 6.2.1).

5.1.1 *Levels A, B, and C.* The cotton cloth put up on rolls shall be packaged in accordance with the applicable requirements of PPP-P-51 for class 4 fabrics.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2 and 6.2.1).

5.2.1 *Levels A, B, and C.* The cotton cloth shall be packed in accordance with the applicable requirements of PPP-P-51 for class 4 fabrics.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-51.

6. NOTES

6.1 Intended use.

6.1.1 *Types I and II.*

6.1.1.1 *Class* 1. Cotton cloth treated for

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fire, water, weather, and mildew resistance as covered by this specification for class 1 is intended for use in the manufacture of canvas covers, tents, tarpaulins, and other duck items providing protection under conditions of prolonged outdoor use.

6.1.1.2 *Class* 2. Cotton cloth treated for fire, water, weather, and mildew resistance as covered by this specification for class 2 is intended for use as upholstery material in motorized vehicles procured by Ordnance Department for the Department of the Army and uses where a higher degree of resistance to crocking is required by the condition of the end use.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Base cloth required (see 3.4.1 and 6.8).

(c) Type and class required (see 1.2.1 and 3.5.5).

(d) Width of cloth required (see 3.5.2 and 6.8).

(e) Bid sample, or laboratory report when not required (see 3.2).

(f) Submission of certification as to the place of finishing, when required (see 3:2).

(g) Flexibility requirements for duck other than numbered or Army duck specified in 3.5.7.

(h) Mildew inhibitor (if other than specified in 3.5.10.1).

(i) Civil Agencies exceptions (see 4.3 and 4.4).

(j) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2.1 In the preparation of contract or orders, it must be noted that various options, choices and alternatives, as indicated in PPP-P-51 may be exercised in the preparation for the delivery of cloth.

6.3 **Accelerated aging.** Where the accelerated aging apparatus specified in 4.4.3 is not available, other tests and requirements may be substituted provided data is submitted showing correlation with the accelerated weathering test of table VI.

6.4 **Standard sample.** For access to standard sample, address the procuring office issuing the invitation for bids.

6.5 **Mildew inhibitors.** The types of mildew-inhibiting agents specified for type I cloth were selected after lengthy and much more severe tests than those set forth in the specification. The specified test is only for the purpose of determining whether this treatment has been applied properly and is not to be considered as a performance test for the valuation of the effectiveness of the mildew-inhibiting treatment. Consideration will be given to the use of other mildew inhibiting agents or treatments which show good resistance to microbiological activity, a low degree of toxicity, and excellent retention of their efficiency following prolonged outdoor exposure, and are acceptable from a toxicological standpoint. At the time of submission of samples to the laboratory, a statement as to the kind of inhibitor used should be included in order to facilitate laboratory testing. Prior to procurement by the contracting activity, the chemical formulations of fire, water, weather, and mildew resistance compounds for all shall be reviewed from a skin toxicity standpoint and approval obtained from the office of the Surgeon General.

6.6 **Gas composition.** A gas mixture known as Matheson Manufactured Gas B, which can be purchased from the Matheson Company, Inc., East Rutherford, New Jersey, conforms to requirements for the synthetic gas mixture specified in table VI.

6.7 **Formula disapproval.** For the information of the contractor, the use of proteins and their derivatives, starch, hydrophylic ma-

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TABLE VII. *Variation in width*

Grey cloth width	Nominal decrease in width due to finishing	Material
34 inches maximum	½ inch	Plied yarn, Army and numbered Ducks
Over 34 inches	1 inch	Plied yarn, Army and numbered Ducks
Up to 40 inches	1-½ inches	Flat ducks requiring desizing
Over 40 inches	2 inches	Flat ducks requiring desizing

terials, oxidizing oils and oxidizing resins and resin or compounds to increase hygroscopicity may lead to formula disapproval (see 3.3 and 3.3.1).

6.8 For the information of procurement officers it is pointed out that applications of this finish affects a change in width from grey to finished state depending upon the width of material and the type of processing required. For guidance, table VII may be used.

6.9 It is believed that Dark Green FWWMR Duck may be finished under the requirements for type I of this specification and the Duck (dark) Blue FWWMR Duck may be finished under the requirements for type II of this specification.

6.10 The transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Cloth, cotton, fireproofed, oiled or waterproofed, not coated, not otherwise indexed by name.

Carload minimum weight—Any quantity.

Motor:

Cloth, cotton, fireproofed, oiled or waterproofed, not coated, not otherwise indexed.

Motor volume minimum weight 30,000 pounds.

6.11 Level B packaging and packing shall be used by ordering agencies only when shipment is known to be made to a holding supply depot or warehouse from which partial shipments will be made to ultimate consumers.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:

Army—M O Q Sig C

Navy—S MC Sh Y Md

Air Force—ASA—MAA

Copies of this specification may be purchased for 15 cents each.

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